INCH-POUND
MIL-DTL-16377/59B(SH)
26 May 2023
SUPERSEDING
MIL-F-16377/59A(SH)
9 May 1983

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT AND SOLID-STATE EQUIVALENT, DETAIL LIGHTING FOR STEP ILLUMINATION, 8 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOLS 353, 353.1, 353.2, 353SSL, 353.1SSL, AND 353.2SSL

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

Unless otherwise noted, all dimensions are in inches. For tolerances other than those specified herein, see 3.5.2 of MIL-DTL-16377.

Figures 1 through 7 provide information for the light fixtures covered by this specification sheet. See the requirements sections for specific references to these figures.

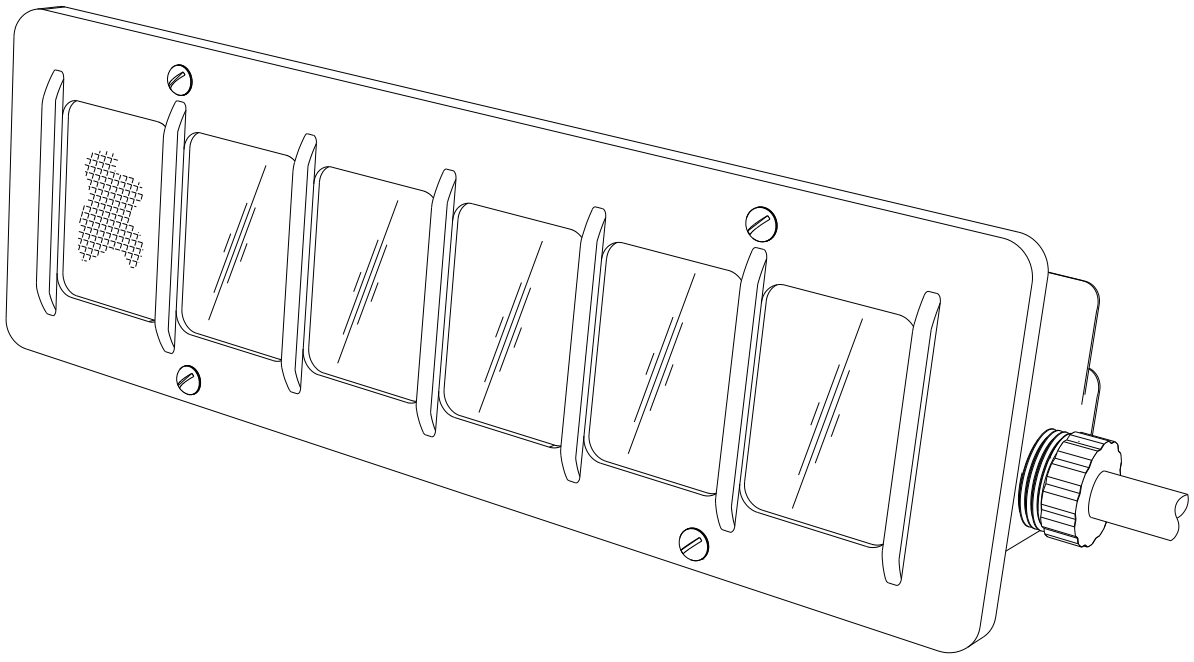


FIGURE 1. Typical view of step illumination fixture.

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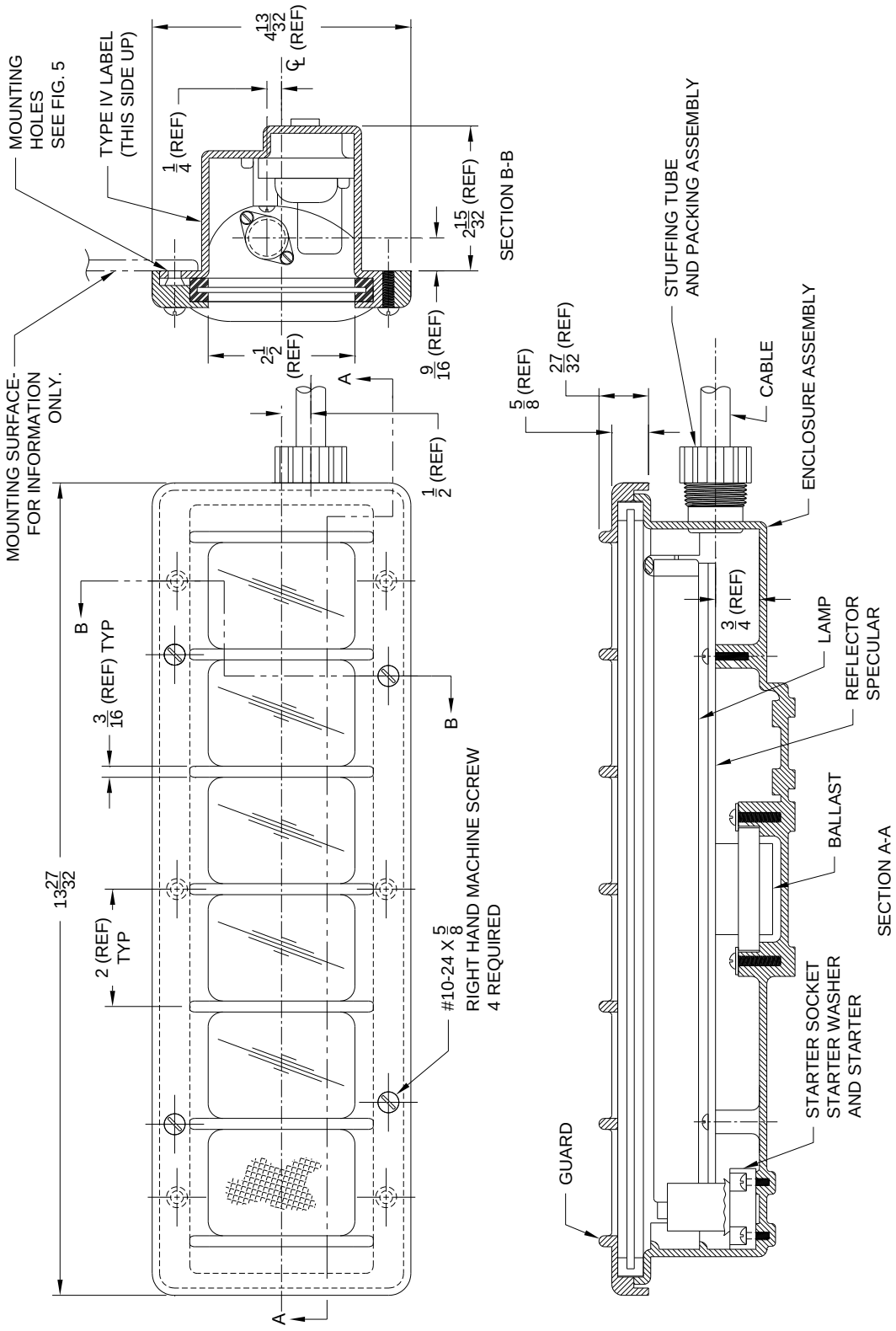
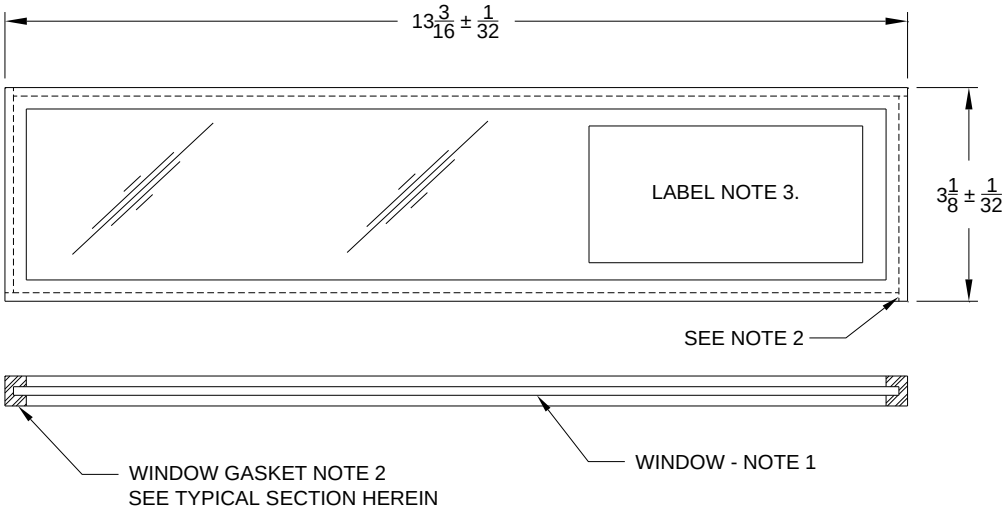
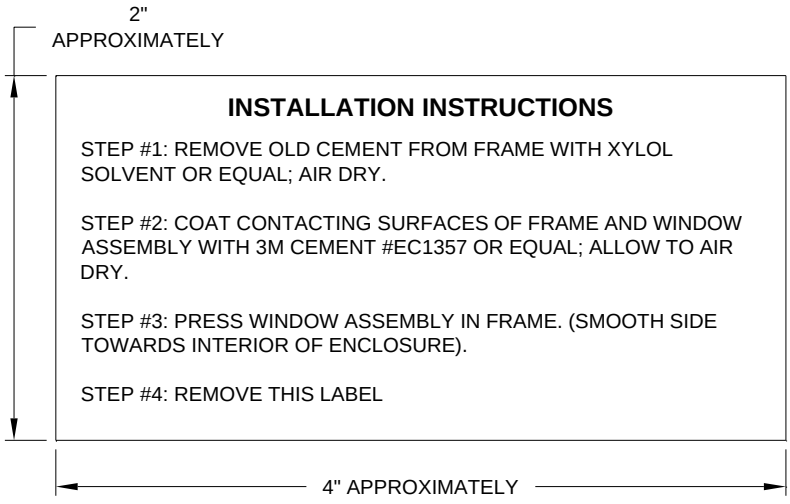


FIGURE 2. Typical views.

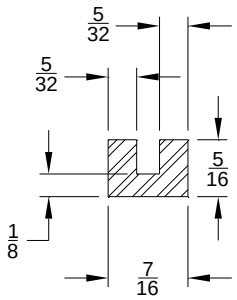
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(a) WINDOW ASSEMBLY



(b) LABEL - NOTE 3.



STANDARD RMA TOLERANCES FOR EXTRUSION SHALL APPLY

(C) GASKET - TYPICAL SECTION.

NOTES:

1. Material for window shall be acrylic plastic as follows:
   Part number M16377/59-001 for symbol 353: Clear, prismatic (one side only), ⅛-inch thick, type CLEAR-2 of MIL-DTL-24191.
   Part number M16377/59-002 for symbol 353.1: Red, prismatic (one side only), ⅛-inch thick, type RTPS-2 of MIL-DTL-24191.
   Part number M16377/59-003 for symbol 353.2: Blue, prismatic (one-side only), ⅛-inch thick, type BTPS-2 of MIL-DTL-24191.
2. Material for window gasket shall be neoprene rubber, 25-35 durometer. Gasket shall be cemented to window. Gasket may be furnished in segments.
3. For window assemblies furnished as repair parts, a type IV label shall be furnished in accordance with MIL-DTL-16377. Location and the information to be shown on this label shall be shown hereon.
4. Window assembly shall be cemented to window frame.

FIGURE 3. Window assembly.

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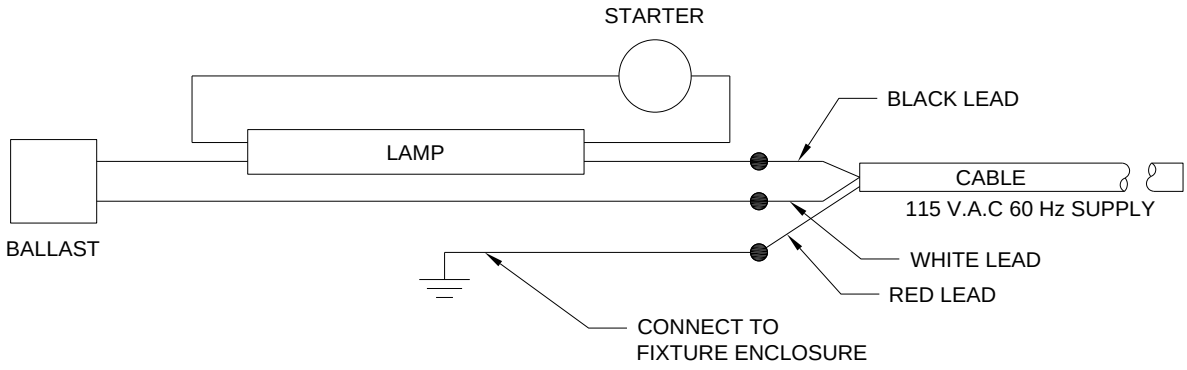
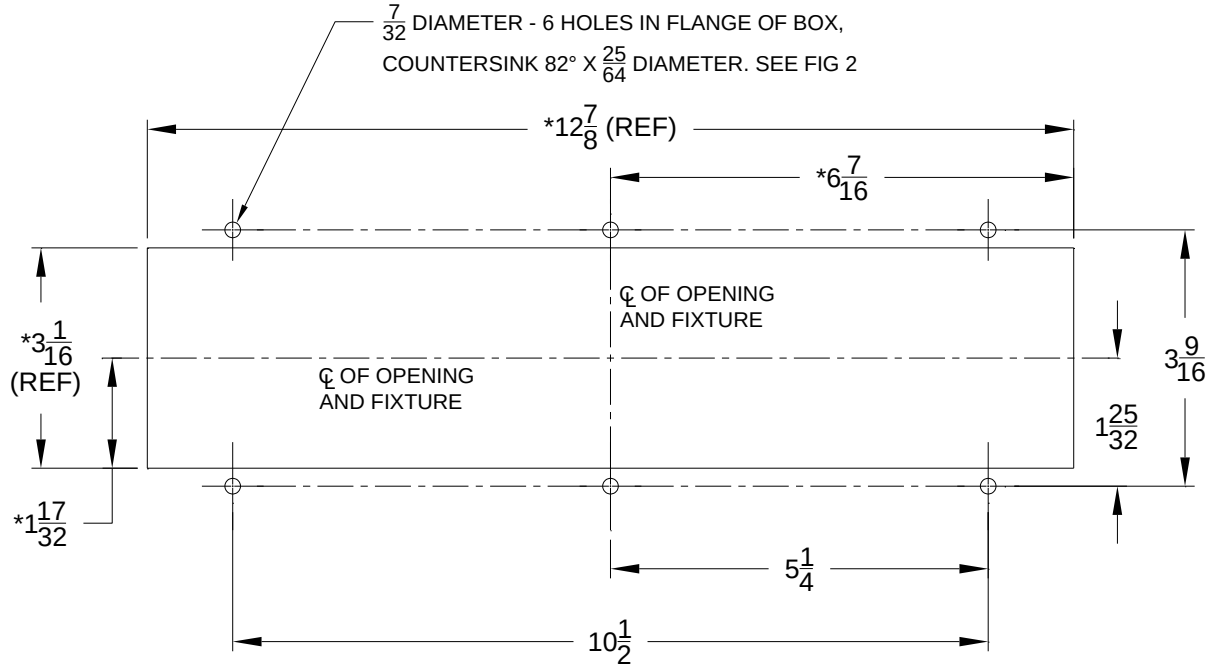


FIGURE 4. Wiring diagram.

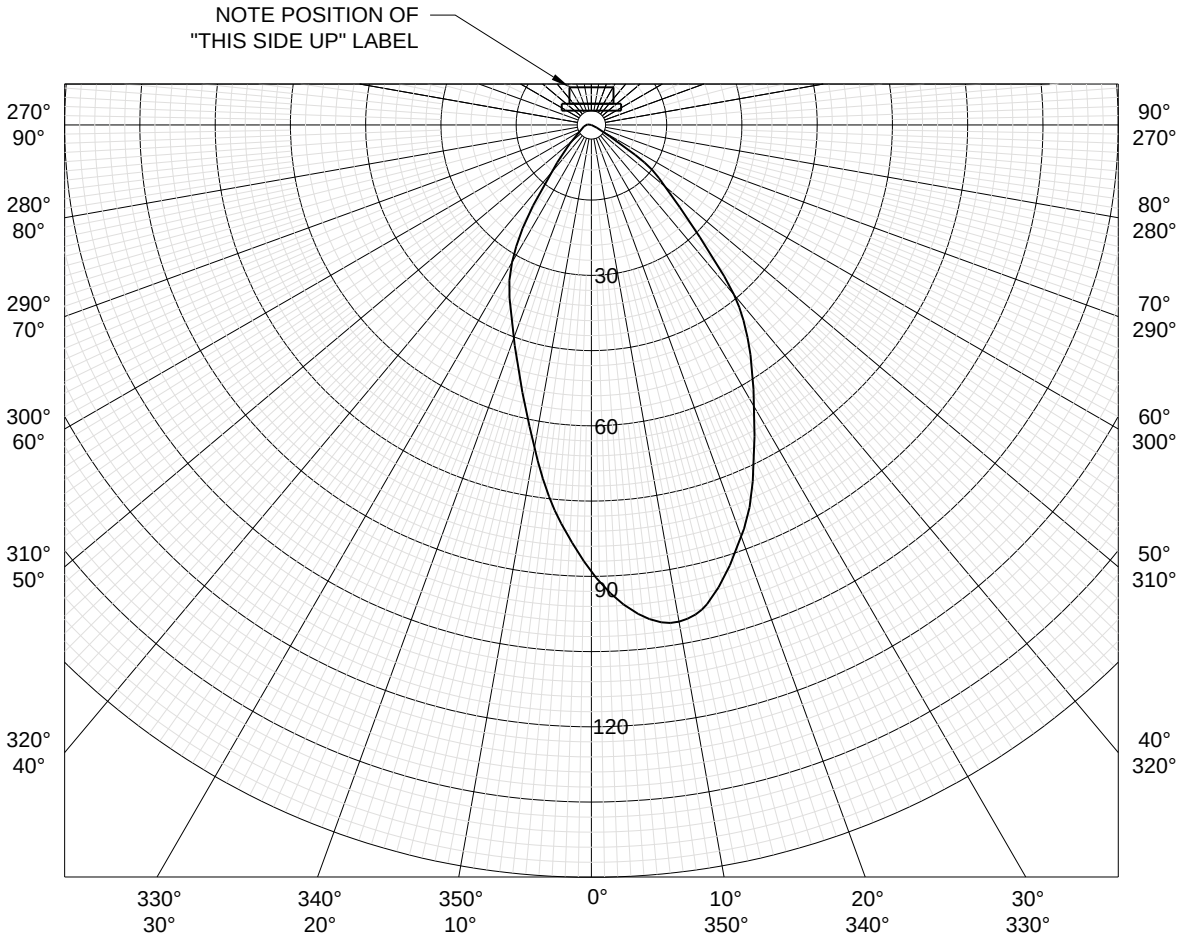


* DIMENSIONS SHOWN ARE FOR OPENING IN MOUNTING SURFACE

MOUNTING BOLTS SHALL BE #10-24 FLAT HEAD MACHINE SCREWS (TO BE FURNISHED BY INSTALLING ACTIVITY.)

FIGURE 5. Mounting dimensions.

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Photometric test data:

Distance: 10 feet

Plane: At right angles to long axis of fixture through geometric center of lamp

Voltage: 115 volts, 60 hertz

Ambient temperature: 76 °F

Lamps: Use lamp furnished with fixtures seasoned for at least 100 hours

Ballast: Use ballast furnished with fixture

Lamp Correction: Distribution curves based on 380 lumens per lamp

FIGURE 6. Candlepower distribution curves.

REQUIREMENTS

CLASSIFICATION

Type: I

Class: 1

SYMBOL NUMBER

353: Complete lighting fixture with one 8-watt cool white lamp and clear prismatic lens assembly.

353.1 Complete lighting fixture with one 8-watt red lamp and clear prismatic lens assembly.

353.2 Complete lighting fixture with one 8-watt blue lamp and clear prismatic lens assembly.

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DESIGN

Electrical rating:	8 watts, 120 volts, 60 hertz.
Dimensions and configuration:	See figures 1 and 2.
Cable:	10 feet of type LSTHOF-3 in accordance with MIL-DTL-24643/3.
Window assembly:	See figure 3.
Wiring diagram:	See figure 4.
Mounting dimensions:	See figure 5.
Light output:	See figure 6 (candlepower).
Enclosure:	Watertight. The enclosure shall be made of 3/32-inch minimum thickness cast aluminum.
Weight:	5 pounds maximum.
Labels:	Labels shall be furnished in accordance with the following types of MIL-DTL-16377: (a) Type I. (b) Type IV. One type IV label shall be applied as shown on figure 2. The information to be shown on the type IV label shall be:"THIS SIDE UP".
Standard parts:	The standard parts shown in table I shall be used in the construction of the fixtures specified herein.

TABLE I. Standard parts.

Name	Specification or Military Part Number	National Stock Number (NSN)	Remarks
Lamp	---	6240-00-299-5546	Industry number F8T5/CW
Lamplock	M16377/40-001	6250-00-399-7039	---
Lampholder	UL 496	6250-00-377-8590	Supplement SB, miniature bipin
Starter socket	UL 496	6250-00-295-2738	Supplement SA, medium bipin
Starter	8W, ANSI C78.180, type III	6250-00-299-5962	Commercial designation: FS5
Ballast	M16377/45-001	6250-00-295-2735	---
Red filter tube	M16377/42-001	6210-00-399-6297	---
Blue filter tube	M16377/42-002	---	---

SOLID-STATE LIGHTING (SSL) REQUIREMENTS

CLASSIFICATION

Type: III
Class: 1

SYMBOL NUMBERS

353SSL: Complete lighting fixture, white.
353.1SSL: Complete lighting fixture, red.
353.2SSL: Complete lighting fixture, blue.

DESIGN

Electrical rating:	8 watts maximum, 120 volts, 60 hertz.
Dimensions:	See figure 2.
Light output:	See figure 6 (candlepower).

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Cable:	10 feet of type LSFHOF-3 in accordance with MIL-DTL-24643/3.
Enclosure:	Watertight.
Chromaticity:	See 3.7.4 of MIL-DTL-16377.
Fail-safe circuit:	See 3.17 of MIL-DTL-16377.
Wiring:	Wiring shall maintain same control scheme as legacy fluorescent fixture.
Weight:	5 pounds maximum.
Designation and marking:	One type I label shall be furnished in accordance with MIL-DTL-16377.

EXAMINATIONS AND TESTS

Examinations and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection:	
Complete fixtures:	See table II for type I and table IV for type III.
Window assemblies:	See table III.
Conformance inspection:	
Comparison inspection:	Same as the first article inspection.
Conformance inspections:	See MIL-DTL-16377.

GENERAL INFORMATION

(a) Assignment of military part numbers and general information is shown in table V.

(b) This light is intended for standing illumination in the berthing areas (red), Combat Information Control spaces (blue), and at the foot of ladders.

TABLE II. First article and comparison inspection for complete fixtures.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Shock	3.5.12	4.8.7	Grade B
Vibration	3.5.13	4.8.8	---
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	---
Insulation resistance	3.6.16	4.8.3	---
Noise	3.5.14	4.8.9	---
Grounding circuit	3.6.13.3	4.8.11	---
Optical uniformity	3.7.4.1.1	4.8.4	---
Light output	3.7.1	4.8.6.1	Candlepower
Magnetic permeability	3.4.8	4.8.18	---

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TABLE III. First article and comparison inspection for window assemblies.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377
Examination	3.12	4.5
Optical uniformity	3.7.4.1.1	4.8.4

TABLE IV. First article and comparison inspection for type III luminaires.

Tests	Requirement Paragraph of MIL-DTL-16377	Test Paragraph of MIL-DTL-16377	Remarks
Examination	3.12	4.5	---
Operation	3.12	4.8.1	---
Power interface	3.6.18	4.8.21	---
Electromagnetic interference	3.6.14	4.8.16	---
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.16	4.8.3	1/
Enclosure effectiveness	3.5.11	4.8.14	Watertight
Shock	3.5.12	4.8.7	Grade B
Vibration	3.5.13	4.8.8	1/
Enclosure effectiveness	3.5.11	4.8.14	Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	1/
Insulation resistance	3.6.14	4.8.3	1/
Noise	3.5.14	4.8.9	1/
Photometrics	3.7.1	4.8.6	Candlepower
Brightness	3.7.5	4.8.6.4	---
Chromaticity	3.7.4	4.8.6.5	---
Correlated color temperature	3.7.4.1.1	4.8.6.6	---
Lumen maintenance	3.7.11	4.8.20	---
Fail-safe circuit design	3.17	4.8.24	---
Optical uniformity	3.7.7.1	4.8.4	---
Stress relief	3.8.5.3	4.8.5	---
Magnetic permeability	3.4.8	4.8.18	---
Grounding circuit	3.6.13.3	4.8.11	---

NOTE:

1/ In accordance with 4.3.1 of MIL-DTL-16377, tests shall be performed on one luminaire in the order specified.

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TABLE V. Part number and general information.

Military Part Number M16377/59-	Item Name	Remarks
353	Complete fixture, symbol 353	---
353.1	Complete fixture, symbol 353.1	---
353.2	Complete fixture, symbol 353.2	---
353SSL	Complete fixture, symbol 353SSL	---
353.1SSL	Complete fixture, symbol 353.1SSL	---
353.2SSL	Complete fixture, symbol 353.2SSL	---
001	Window assembly	See figure 3

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 6210-2023-017)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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